ESE 101:

Homework 2 (due October 23, 2025):

Upload your full code in addition to answers to the following questions to receive full credit.

0.1 Jupiter

Jupiter's mean distance from the sun is 7.8×10^8 km (or 5.2 Astronomical Units). Its (Bond) albedo is 0.34, and the solar luminosity is 3.8×10^{26} W.

1. What is the total solar irradiance at Jupiter?
2. In your own words (no math), explain brightness temperature and effective temperature to your college-level friend who is not taking this class. Make sure to mention the relationship between the two of them. Feel free to add drawings if useful to your explanation.
3. What would be the effective temperature of Jupiter if it were a perfect blackbody (in the longwave band) in equilibrium with the insolation?
4. The measured effective temperature of Jupiter is 125 K. How large is the additional energy flux that must be emanating from Jupiter's interior to lead to this effective temperature? *[This energy flux is somewhat analogous to the (small) geothermal energy flux on Earth, but it is much larger and arises through different processes: It arises primarily because Jupiter is still contracting, and the gravitational energy that is released by the contraction emanates as an internal energy flux.]*

(*40 points total - 10 each*)

0.2 Insolation

Write a program that calculates the diurnally averaged insolation as a function of time t given the orbital parameters obliquity γ, longitude of perihelion ϖ, and eccentricity e. (See the book draft for how to calculate insolation.) If you're interested in higher order expansions or variations in orbital parameters consider referencing the source code used in the CliMA earth system model (`https://github.com/CliMA/Insolation.jl/`). With your program, please do the following:

1. Calculate and plot the insolation as function of latitude at the top of Earth's atmosphere for each day of 2025.
2. Calculate and plot the annually averaged insolation as a function of latitude.
3. Turn the longitude of perihelion 180° compared to its present value. What is the date of perihelion now? Plot insolation for each day of year and averaged over the year. Make a plot of the differences between the insolation for each day of the year. What are the principal differences to the situations in parts a and b? How do they arise?
4. Now reset the longitude of perihelion to its present value and decrease the obliquity to 20°. Make a plot of the differences. What are the principal differences to the situation in parts a and b now, and how do they arise?
5. Earth's longitude of perihelion and obliquity vary almost periodically, with periods of about 21 kyr and 40 kyr. Given your results in the previous two exercises, how do you think can precession of the longitude of perihelion and/or obliquity variations lead to the inception and termination of glacial periods? The following paper is a fascinating discussion of ice ages as non-linear oscillators paced by Milankovich cycles and has some possible answers ;) `https://doi.org/10.1029/2005PA001241`
6. Repeat part 4 (same longitude of perhelion as present, different obliquity) with an obliquity of 60°. What new phenomenon arises and why?
7. Uranus has an obliquity of 97.86°. Where is the annually averaged insolation maximal on Uranus?

(*60 points - 10 each for 1-5, 5 for 6,7*)